

HSC Practice Questions

Investments

- 1 Peter currently earns \$21.50 per hour. His hourly wage will increase by 2.1% compounded each year for the next four years.

What will his hourly wage be after four years?

- A $21.50 (1.21)^4$
 B $21.50 (1.021)^4$
 C $21.50 + 21.50 \times 0.21 \times 4$
 D $21.50 + 21.50 \times 0.021 \times 4$ (1 mark)

(Q5, 2021 HSC) Easy

- 2 Nina plans to invest \$35 000 for 1 year. She is offered two different investment options.

Option A: Interest is paid at 6% per annum compounded monthly.

Option B: Interest is paid at $r\%$ per annum simple interest.

- a Calculate the future value of Nina's investment after 1 year if she chooses Option A. (2 marks) Medium
 b Find the value of r in Option B that would give Nina the same future value after 1 year as for Option A. Give your answer correct to two decimal places. (2 marks) Hard

(Q26, 2021 HSC)

- 3 Ariana owns 1500 shares in a company. The market price for each share is \$27. Ariana's total dividend from these shares is \$810.

Calculate the dividend yield for her shares.

(2 marks)

(Q30, 2021 HSC) Medium

- 4 Joan invests \$200. She earns interest at 3% per annum, compounded monthly. What is the future value of Joan's investment after 1.5 years?

- A \$209.07 B \$209.19
 C \$279.51 D \$311.93 (1 mark)

(Q4, 2020 HSC) Medium

- 5 The inflation rate over the year from January 2019 to January 2020 was 2%.

The cost of a school jumper in January 2020 was \$122.

Calculate the cost of the jumper in January 2019 assuming that the only change in the cost of the jumper was due to inflation. (2 marks)

(Q21, 2020 HSC) Medium

- 6 Jana owns a share portfolio. Details of her share portfolio at 30 June 2020 are given in the table.

Company name	Number of shares in Jana's portfolio	Dividend yield (per annum)	Market price per share
ABC	200	6.0%	\$5.50
XYZ	?	4.0%	\$6.00

Jana received a total annual dividend of \$149.52 from her share portfolio.

Calculate the number of shares Jana has in company XYZ on 30 June 2020. (3 marks)

(Q29, 2020 HSC) Medium

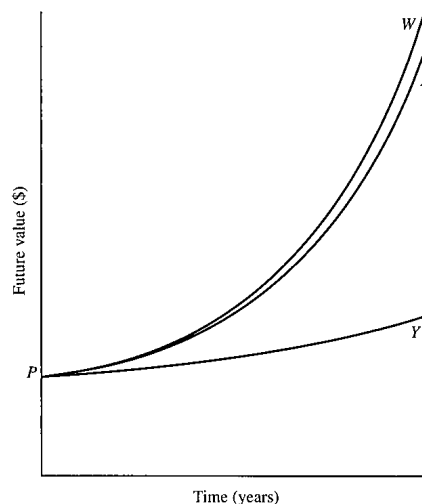
- 7 Chris opens a bank account and deposits \$1000 into it. Interest is paid at 3.5% per annum, compounding annually.

Assuming no further deposits or withdrawals are made, what will be the balance in the account at the end of two years?

- A \$1070.00 B \$1071.23
 C \$1822.50 D \$2070.00 (1 mark)

(Q3, 2019 HSC) Easy

- 8 The graphs show the future values over time of \$P, invested at three different rates of compound interest.



Which of the following correctly identifies each graph?

A

W	5% pa, compounding annually
X	10% pa, compounding annually
Y	10% pa, compounding quarterly

B

W	5% pa, compounding annually
X	10% pa, compounding quarterly
Y	10% pa, compounding annually

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C

W	10% pa, compounding quarterly
X	10% pa, compounding annually
Y	5% pa, compounding annually

D

W	10% pa, compounding annually
X	10% pa, compounding quarterly
Y	5% pa, compounding annually

(1 mark)

(Q13, 2019 HSC) **Medium**

- 9 A person owns 1526 shares with a market value of \$8.75 per share. The total dividend received for these shares is \$1068.20.

Calculate the percentage dividend yield.

(2 marks)

(Q21, 2019 HSC) **Medium**

- 10 Lucas invests \$80 000 for 5 years, at an interest rate of 6% per annum, compounded annually. Ethan invests \$2500 at the end of each month for 5 years, at an interest rate of 6% per annum, compounded monthly. Financial gain is defined as the difference between the final value of an investment and the total contribution. Who will have the better financial gain after 5 years, and by what amount? Use the table below and justify your answer with suitable calculations.

Period	Table of future value interest factors							
	Interest rate per period							
	0.50%	1.00%	1.50%	2.00%	3.00%	4.00%	5.00%	6.00%
5	5.0503	5.1010	5.1523	5.2040	5.3091	5.4163	5.5256	5.6371
6	6.0755	6.1520	6.2296	6.3081	6.4684	6.6330	6.8019	6.9753
7	7.1059	7.2135	7.3230	7.4343	7.6625	7.8983	8.1420	8.3938
8	8.1414	8.2857	8.4328	8.5830	8.8923	9.2142	9.5491	9.8975
12	12.3356	12.6825	13.0412	13.4121	14.1920	15.0258	15.9171	16.8699
24	25.4320	26.9735	28.6335	30.4219	34.4265	39.0826	44.5020	50.8156
36	39.3361	43.0769	47.2760	51.9944	63.2759	77.5983	95.8363	119.1209
48	54.0978	61.2226	69.5652	79.3535	104.4084	139.2632	188.0254	256.5645
60	69.7700	81.6697	96.2147	114.0515	163.0534	237.9907	353.5837	533.1282
72	86.4089	104.7099	128.0772	158.0570	246.6672	396.0566	650.9027	1089.6286

(3 marks)

Bonus question (see page iv) **Hard**

- 11 A company with a share price of \$3.42 declares a dividend of 14 cents. What is the dividend yield as a percentage, correct to two decimal places?

(1 mark)

Bonus question **Medium**

- A $21\,000 \times 1.0816$ B $21\,000 \div 1.0816$
C $21\,000 \times 1.0829$ D $21\,000 \div 1.0829$

(1 mark)

(Q19, 2018 HSC) **Medium**

- 12 The table shows the compounded values of \$1 at different interest rates over different periods.

Compounded values of \$1

Number of periods	Interest rate per period				
	1%	2%	3%	4%	5%
2	1.0201	1.0404	1.0609	1.0816	1.1025
4	1.0406	1.0824	1.1255	1.1699	1.2155
6	1.0615	1.1262	1.1941	1.2653	1.3401
8	1.0829	1.1717	1.2668	1.3686	1.4775
10	1.1046	1.2190	1.3439	1.4802	1.6289
12	1.1268	1.2682	1.4258	1.6010	1.7959

Amy hopes to have \$21 000 in 2 years to buy a car. She opens an account today which pays interest of 4% pa, compounded quarterly.

Using the table, which expression calculates the minimum single sum that Amy needs to invest today to ensure she reaches her savings goal?

- 13 A single amount of \$10 000 is invested for 4 years, earning interest at the rate of 3% per annum, compounded monthly.

Which expression will give the future value of the investment?

- A $10\,000 \times (1 + 0.03)^4$
B $10\,000 \times (1 + 0.03)^{48}$
C $10\,000 \times \left(1 + \frac{0.03}{12}\right)^4$
D $10\,000 \times \left(1 + \frac{0.03}{12}\right)^{48}$

(1 mark)

(Q10, 2017 HSC) **Easy**

- 14 Sam purchased 500 company shares at \$3.20 per share. Brokerage fees were 1.5% of the purchase price.

Sam is paid a dividend of 26 cents per share, then immediately sells the shares for \$4.80 each.

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If he pays no further brokerage fees, what is Sam's total profit? (3 marks)

(Q26e, 2017 HSC) **Medium**

- 15 The table shows the future value of an investment of \$1000, compounding yearly, at varying interest rates for different periods of time.

Future values of an investment of \$1000

Number of years	Interest rate per annum				
	1%	2%	3%	4%	5%
1	1010.00	1020.00	1030.00	1040.00	1050.00
2	1020.10	1040.40	1060.90	1081.60	1102.50
3	1030.30	1061.21	1092.73	1124.86	1157.63
4	1040.60	1082.43	1125.51	1169.86	1215.51
5	1051.01	1104.08	1159.27	1216.65	1276.28

Based on the information provided, what is the future value of an investment of \$2500 over 3 years at 4% pa?

- A \$1124.86 B \$2812.15
C \$3624.86 D \$5312.15 (1 mark)

(Q8, 2016 HSC) **Easy**

- 16 What amount must be invested now at 4% per annum, compounded quarterly, so that in five years it will have grown to \$60 000?

- A \$8919 B \$11 156
C \$49 173 D \$49 316 (1 mark)

(Q17, 2015 HSC) **Easy**

- 17 A family currently pays \$320 for some groceries.

Assuming a constant annual inflation rate of 2.9%, calculate how much would be paid for the same groceries in 5 years' time. (2 marks)

(Q26d, 2015 HSC) **Easy**

- 18 Chandra and Sascha plan to have \$20 000 in an investment account in 15 years time for their grandchild's university fees.

The interest rate for the investment account will be fixed at 3% per annum compounded monthly.

Calculate the amount that they will need to deposit into the account now in order to achieve their plan. (3 marks)

(Q30a, 2014 HSC) **Medium**

- 19 Kimberley has invested \$3500.

Interest is compounded half-yearly at a rate of 2% per half-year.

Compounded values of \$1

Period	Interest rate per period					
	1%	2%	3%	4%	5%	6%
1	1.010	1.02	1.03	1.04	1.05	1.06
2	1.020	1.040	1.061	1.082	1.103	1.124
3	1.030	1.061	1.093	1.125	1.158	1.191
4	1.041	1.082	1.126	1.170	1.216	1.262
5	1.051	1.104	1.159	1.217	1.276	1.338
6	1.062	1.126	1.194	1.265	1.340	1.419
7	1.072	1.149	1.230	1.316	1.407	1.504
8	1.083	1.172	1.267	1.369	1.477	1.594

Use the table to calculate the value of her investment at the end of 4 years. (2 marks)

(Q26e, 2013 HSC) **Medium**

- 20 Adhele has 2000 shares. The current share price is \$1.50 per share. Adhele is paid a dividend of \$0.30 per share.

- i What is the current value of her shares? (1 mark) **Easy**
ii Calculate the dividend yield. (1 mark) **Medium**

(Q28d, 2013 HSC)

- 21 Tracy invests some money for 2 years at 4% per annum, compounded quarterly.

Compounded values of \$1

Period	Interest rate per period				
	1%	2%	3%	4%	5%
1	1.010	1.020	1.030	1.040	1.050
2	1.020	1.040	1.061	1.082	1.103
3	1.030	1.061	1.093	1.125	1.158
4	1.041	1.082	1.126	1.170	1.216
5	1.051	1.104	1.159	1.217	1.276
6	1.062	1.126	1.194	1.265	1.340
7	1.072	1.149	1.230	1.316	1.407
8	1.083	1.172	1.267	1.369	1.477

Which figure from the table should Tracy use to calculate the value of her investment at the end of 2 years?

- A 1.020 B 1.082
C 1.083 D 1.369 (1 mark)

(Q9, 2012 HSC) **Medium**

- 22 A house was purchased in 1984 for \$35 000. Assume that the value of the house has increased by 3% per annum since then.

Which expression gives the value of the house in 2009?

- A $35\,000(1 + 0.03)^{25}$ B $35\,000(1 + 3)^{25}$
C $35\,000 \times 25 \times 0.03$ D $35\,000 \times 25 \times 3$ (1 mark)

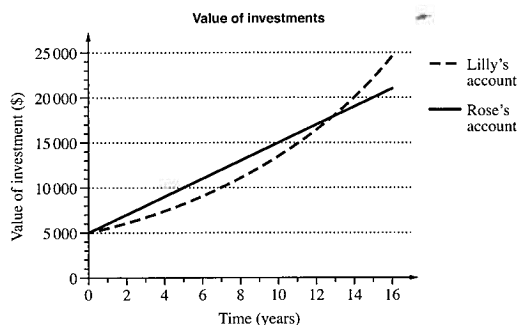
(Q6, 2009 HSC) **Easy**

- 23 Lilly and Rose each have money to invest and choose different investment accounts.

The graph shows the values of their investments over time.

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- i How much was Rose's original investment? (1 mark) **Easy**
- ii At the end of 6 years, which investment will be worth the most and by how much? (2 marks) **Medium**
- iii Lilly's investment will reach a value of \$20 000 first. How much longer will it take Rose's investment to reach a value of \$20 000? (1 mark) **Medium**

(Q23a, 2007 HSC)

- 24 Last year, Helen bought 150 shares at \$2.00 per share. They are now worth \$2.50 per share. Helen receives a dividend of \$0.10 per share.

What is the dividend yield?

- A 4% B 20%
C \$15 D \$75 (1 mark)

(Q13, 2005 HSC) **Easy**

- 25 Minh invests \$24 000 at an interest rate of 4.75% per annum, compounded monthly. What is the value of the investment after 3 years? (2 marks)

(Q24a, 2003 HSC) **Medium**

- 26 Richard has 2000 shares with a current market value of \$4.80 each. During the past twelve months, Richard received a total dividend of \$240.

What is the current dividend yield on these shares?

- A 0.025% B 2%
C 2.5% D 40% (1 mark)

(Q7, 2002 HSC) **Medium**

- 27 i Katherine invests \$50 000 with Standard Credit Union for a term of 5 years. Her investment earns interest at 3.1% per annum compounded annually. How much will Katherine's investment be worth at the end of 5 years? Give your answer to the nearest dollar. (2 marks) **Easy**

- ii Katherine's sister Liz also has \$50 000 to invest for a term of 5 years. She invests with General Bank. Her investment earns interest at 3% per annum, compounded monthly. Which sister makes the better investment? Justify your answer with appropriate calculations. (2 marks) **Medium**

(Q23b, 2002 HSC)

Investments

1 $PV = \$21.50, r = 0.021, n = 4$

$$FV = PV(1 + r)^n$$

$$= \$21.50(1 + 0.021)^4$$

$$= \$21.50(1.021)^4$$

Answer B

2 a Option A:

$$PV = \$35\,000,$$

$$r = \frac{0.06}{12} = 0.005, n = 12 \quad \checkmark$$

$$FV = PV(1 + r)^n$$

$$= \$35\,000(1 + 0.005)^{12}$$

$$= \$37\,158.7234...$$

$$= \$37\,158.72 \quad (\text{nearest cent}) \quad \checkmark$$

(2 marks)

b Interest

$$= \$37\,158.72 - \$35\,000$$

$$= \$2\,158.72$$

Option B:

$$I = Prn$$

$$\$2\,158.72 = \$35\,000 \times \frac{r}{100} \times 1 \quad \checkmark$$

$$2158.72 = 350r$$

$$r = \frac{2158.72}{350}$$

$$= 6.167\,771\,42...$$

$$= 6.17 \quad (2 \text{ d.p.}) \quad \checkmark$$

(2 marks)

3 Dividend per share

$$= \$810 \div 1500$$

$$= \$0.54 \quad \checkmark$$

$$\text{Dividend yield} = \frac{\$0.54}{\$27} \times 100\%$$

$$= 2\% \quad \checkmark$$

(2 marks)

4 $FV = PV(1 + r)^n$

$$= \$200 \left(1 + \frac{0.03}{12}\right)^{18}$$

$$= \$209.193\,824...$$

$$= \$209.19 \quad (\text{nearest cent})$$

Answer B

5 $PV = \frac{FV}{(1 + r)^n}$

$$= \frac{\$122}{(1 + 0.02)^1}$$

$$= \$119.607\,843...$$

$$= \$119.61 \quad (\text{nearest cent}) \quad \checkmark$$

The cost of the jumper in 2019 was \$119.61 to the nearest cent. $\checkmark$

(2 marks)

6

Company name	Number of shares in Jana's portfolio	Dividend yield (per annum)	Market price per share
ABC	200	6.0%	\$5.50
XYZ	?	4.0%	\$6.00

Total value of ABC shares

$$= 200 \times \$5.50$$

$$= \$1100$$

$$\text{Dividend} = 6\% \text{ of } \$1100$$

$$= \$66.00 \quad \checkmark$$

Dividend from XYZ shares

$$= \$149.52 - \$66.00$$

$$= \$83.52 \quad \checkmark$$

So \$83.52 is 4% of their value.

$$\text{Value is } \$83.52 \div 4 \times 100 \text{ or } \$2088.$$

$$\text{Number of shares} = \$2088 \div \$6$$

$$= 348$$

So Jana has 348 shares in XYZ. $\checkmark$

(3 marks)

7 $PV = \$1000, r = 0.035, n = 2$

$$FV = PV(1 + r)^n$$

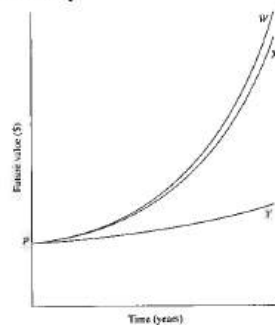
$$= \$1000(1 + 0.035)^2$$

$$= \$1071.225$$

The balance will be \$1071.23.

Answer B

8 The future value will be least at 5% p.a. compounding annually. The future value will be greatest at 10% p.a. compounding quarterly. So Y will be 5% p.a. compounding annually, W will be 10% p.a. compounding quarterly and X 10% p.a. compounding annually.



Answer C

9 Dividend per share

$$= \$1068.20 \div 1526$$

$$= \$0.70 \quad \checkmark$$

Percentage dividend yield

$$= \frac{\$0.70}{\$8.75} \times 100\%$$

$$= 8\% \quad \checkmark$$

(2 marks)

10 Lucas: $FV = PV(1 + r)^n$

$$= 80\,000 \times 1.06^5$$

$$= 107\,058.05 \quad (2 \text{ dec. pl.}) \quad \checkmark$$

Financial gain

$$= 107\,058.05 - 80\,000$$

$$= 27\,058.05$$

$\therefore$ Lucas's gain is \$27 058.05. $\checkmark$

Ethan: 6% p.a.

$$= 0.5\% \text{ per month, for 5 years}$$

$$= 60 \text{ months}$$

$$FV = 2500 \times 69.7700 \text{ (from table)}$$

$$= 174\,425 \quad \checkmark$$

Financial gain

$$= 174\,425 - 2500 \times 60$$

$$= 24\,425$$

$\therefore$ Ethan's gain is \$24 425.

$\therefore$ Lucas's gain is \$2633.05 more than Ethan's gain. $\checkmark$

(3 marks)

11 Dividend yield

$$= \frac{0.14}{3.42} \times 100\%$$

$$= 4.093\,567\,251...$$

$$= 4.09 \quad (2 \text{ dec. pl.}) \quad \checkmark$$

(1 mark)

12 2 years = 2 $\times$ 4 quarters

So the number of periods is 8.

4% p.a. = 1% per quarter

From the table the required factor is 1.0829.

$$\text{So required amount} \times 1.0829$$

$$= \$21\,000$$

$$\text{Amount} = \$ (21\,000 \div 1.0829)$$

Answer D

13 $PV = 10\,000, r = \frac{0.03}{12},$

$$n = 4 \times 12 = 48$$

$$FV = PV(1 + r)^n$$

$$= 10\,000 \times \left(1 + \frac{0.03}{12}\right)^{48}$$

Answer D

14 Price of shares = 500 $\times$ \$3.20

$$= \$1600 \quad \checkmark$$

Brokerage = 1.5% of \$1600

$$= 0.015 \times \$1600$$

$$= \$24$$

Total cost = \$1600 + \$24

$$= \$1624$$

Total dividend = 500 $\times$ \$0.26

$$= \$130 \quad \checkmark$$

Total sales = 500 $\times$ \$4.80

$$= \$2400$$

Total received = \$130 + \$2400

$$= \$2530$$

Profit = \$2530 - \$1624

$$= \$906 \quad \checkmark$$

(3 marks)

15 $FV = \$ (2.5 \times 1124.86)$

$$= \$2812.15$$

Answer B

Investments

$$\begin{aligned}
 16 \quad PV &= \frac{FV}{(1+r)^n} \\
 &= \frac{\$60000}{\left(1 + \frac{0.04}{4}\right)^{5 \times 4}} \\
 &= \$49\,172.6682\dots \\
 &= \$49\,173 \quad (\text{nearest dollar})
 \end{aligned}$$

Answer C

$$\begin{aligned}
 17 \quad FV &= PV(1+r)^n \\
 &= \$320(1+0.029)^5 \\
 &= \$369.170\,38\dots \\
 &= \$369 \quad (\text{nearest dollar})
 \end{aligned}$$

Around \$369 would be paid for the groceries in 5 years' time. ✓
(2 marks)

$$\begin{aligned}
 18 \quad PV &= \frac{FV}{(1+r)^n} \\
 &= \frac{\$20\,000}{\left(1 + \frac{0.03}{12}\right)^{15 \times 12}} \quad \checkmark\checkmark \\
 &= \$12\,759.7264\dots \\
 &= \$12\,760 \quad (\text{nearest dollar}) \quad \checkmark (3 \text{ marks})
 \end{aligned}$$

$$\begin{aligned}
 19 \quad &4 \text{ years is } 8 \text{ half-years so the period is } 8. \\
 &\text{The interest rate per period is } 2\%. \quad \checkmark \\
 &\text{Value of investment} \\
 &= 1.172 \times \$3500 \\
 &= \$4102 \quad \checkmark (2 \text{ marks})
 \end{aligned}$$

$$\begin{aligned}
 20 \quad \text{i} \quad \text{Current value} &= 2000 \times \$1.50 \\
 &= \$3000 \quad (1 \text{ mark}) \\
 \text{ii} \quad \text{Dividend yield} &= \frac{0.30}{1.50} \times 100\% \\
 &= 20\% \quad (1 \text{ mark})
 \end{aligned}$$

$$\begin{aligned}
 21 \quad &4\% \text{ pa} = 1\% \text{ per quarter} \\
 &\text{So the interest rate per period is } 1\%. \\
 &2 \text{ years} = 8 \text{ quarters} \\
 &\text{So the number of periods is } 8. \\
 &\text{From the table, the required figure is } 1.083. \\
 &\text{Answer C}
 \end{aligned}$$

$$\begin{aligned}
 22 \quad A &= P(1+r)^n \\
 \therefore A &= 35000(1+0.03)^{25} \\
 &\text{Answer A}
 \end{aligned}$$

$$\begin{aligned}
 23 \quad \text{i} \quad &\text{Rose's original investment was } \$5000, (\text{when } t = 0). \quad (1 \text{ mark}) \\
 \text{ii} \quad &\text{At the end of 6 years, Rose has } \$11\,000 \text{ and Lilly has } \$9000. \quad \checkmark
 \end{aligned}$$

$\therefore$ Rose's investment is worth the most by \$2000. ✓
(2 marks)

$$\begin{aligned}
 \text{iii} \quad &\text{Lilly's investment takes } 14 \text{ years while Rose's takes } 15 \text{ years.} \\
 &\therefore \text{It will take Rose } 1 \text{ year longer.} \quad (1 \text{ mark})
 \end{aligned}$$

$$\begin{aligned}
 24 \quad \text{Dividend yield} &= \frac{\text{dividend}}{\text{market value}} \times 100 \\
 &= \frac{0.10}{2.50} \times 100 \\
 &= 4\%
 \end{aligned}$$

Answer A

$$\begin{aligned}
 25 \quad A &= P(1+r)^n \\
 P &= \$24\,000 \\
 r &= 4.75\% \text{ p.a.} \\
 &= \frac{4.75}{12} \% \text{ p. month} \\
 &= 0.3958\bar{3} \% \text{ p. month} \\
 n &= 3 \text{ years} \\
 &= 36 \text{ months} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \therefore A &= 24\,000 \left(1 + \frac{0.3958\bar{3}}{100}\right)^{36} \\
 &= \$27\,667.8901\dots \\
 &= \$27\,667.89 \quad \checkmark (2 \text{ marks})
 \end{aligned}$$

$$\begin{aligned}
 26 \quad \text{Total market value} &= 2000 \times \$4.80 \\
 &= \$9600 \\
 \text{Total dividend} &= \$240 \\
 \text{Dividend yield} &= \frac{240}{9600} \times 100 \\
 &= 2.5\%.
 \end{aligned}$$

Answer C

$$\begin{aligned}
 27 \quad \text{i} \quad A &= P(1+r)^n \\
 &= 50000(1+0.031)^5 \quad \checkmark \\
 &= \$58\,245.63 \\
 &= \$58\,246 \quad \checkmark \\
 &\text{to nearest dollar} \quad (2 \text{ marks})
 \end{aligned}$$

$$\begin{aligned}
 \text{ii} \quad A &= P(1+r)^n \\
 P &= \$50\,000 \\
 r &= 3 \div 12 = 0.25\% \text{ per month} \\
 n &= 5 \times 12 = 60 \text{ months} \quad \checkmark \\
 \therefore A &= 50000(1+0.0025)^{60} \\
 &= \$58\,080.84 \\
 &= \$58\,081 \\
 &\text{to nearest dollar} \\
 \therefore &\text{Katherine's investment is marginally better by } \$165. \quad \checkmark (2 \text{ marks})
 \end{aligned}$$